Selene An

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Education

- **University of Maryland - College Park** College Park, MD
  B.S. of Computer Science, B.S. of Mathematics; GPA: 3.5 *Aug 2021 – May 2024*
  - **Coursework**: Web Application Development, Programming Handheld Systems, Computer Graphics, Data Structure, Cryptography, Combinatorics, Graph Theory, Abstract Algebra, Linear Algebra

Technical Skills

- **Languages**: Python, C++, C, Java, JavaScript, HTML, CSS, Dart, OCaml, Rust, Ruby, MATLAB
- **Technologies/Frameworks**: Git, Shell, React, Flutter, Spring Boot, Django, Express, Node.js, SQL, Firebase, Redis, MongoDB, Docker, Amazon Web Services, Linux, WebGL

Experience

- **FlowMingle, Inc.** Remote
  Software Engineer Intern *Mar 2024 - Present*
  - Revamped start page interface for **website**, **iOS**, and **Android** platforms with a focus on modern and user-friendly design. Utilized **Flutter** for cross-platform development and **Figma** for **UI/UX** design.
  - Created a responsive image slider component for the start page to improve visual appeal and user engagement. Ensured smooth performance and seamless transitions across different devices and screen sizes.
  - Developed a robust **backend** service using **Node.js** and **Firebase**, enabling users to generate shareable event preview links, increasing event visibility on social media by 26%.
  - Implemented a secure sign-in/sign-up functionality using **Firebase** Authentication, supporting multi-country phone number validation for global users, improving sign-up rates by 14% across all regions.
  - Added a co-hosting feature to the event management system, enhancing event hosting flexibility and increasing overall user engagement by 18%, especially for multi-organizer events.
  - Implemented event deletion and cancellation features, streamlining event lifecycle management and reducing event management errors by 20% through better data consistency.
  - Redesigned desktop user interfaces by implementing modern, responsive layouts for event pages, improving the user experience on larger screens and driving a 25% in desktop conversions across all major browsers.

Projects

- **Personal Blog Website** *Mar 2024 - May 2024*
  - Designed and developed a scalable blog application with distinct user/admin roles using **Spring Boot**; implemented **RESTful APIs** for CRUD operations with **MySQL**, optimized with pagination.
  - Created detailed data schemas with ER diagrams for user and metadata management. Set up and configured MySQL databases on **AWS** cloud using **RDS** Service for scalable and efficient cloud-based data storage.
  - Leveraged **Spring Data JPA** with **Hibernate** for post-management, ensuring low latency access to **MySQL** database and optimizing performance for high-traffic applications.
  - Implemented user authentication and authorization with **Spring Security OAuth** and **JWT** in **Java**.
  - Containerized the web app with **Docker** and deployed it to the AWS cloud using **Elastic Beanstalk** for scalability.
- **Translation Service Website** *Oct 2023 - Dec 2023*
  - Built a user-friendly translation service using Google Translate API; developed with **Node.js**, **Express**, and **MongoDB** for seamless CRUD operations.
  - Constructed **MongoDB** databases and created data models with **Mongoose** for data validation.
  - Created responsive frontend pages using HTML, CSS, JavaScript, and **Bootstrap**, using **JQuery** to respond to user events, and hooked the backend through **Ajax** requests.

Research

- **Directed Reading Programming: Algebraic Graph Theory** *Feb 2022 - May 2022*
  - Verified the properties of the Ramanujan Graph for points and edges based on the theory of algebraic graph theory.
- **Designing Efficient Algorithms for Domination Problems with Applications** *Jun 2021 - Aug 2021*
  - Analyzed the mathematical properties of domination set in graphs and designed efficient algorithms for computing the minimum-size set of domination for general or special classes of graphs.